

Tools and Weapons: The Promise and the Peril of the Digital Age

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Stelle: 633

Our “cloud commitments”: privacy, security, compliance, and transparency.

Stelle: 636

Developing clear principles and putting them to work were two separate challenges. The team built out each principle with details and training. The real test would come when new circumstances raised tough questions and required us to decide how far we would go to stand up for the commitments we had created.

Stelle: 830

With everyone’s personal information stored in the cloud, an authoritarian regime bent on broad surveillance can unleash draconian demands to monitor not only what people are communicating, but even what they’re reading and watching online. And armed with this knowledge, governments can prosecute, persecute, or even execute those individuals they consider threats.

Stelle: 872

As cloud computing became more ubiquitous and data more accessible, however, the temptation for governments to act unilaterally to seek data in other countries proved irresistible.

Stelle: 992

Clarifying Lawful Overseas Use of Data Act, or CLOUD Act. The legislation had provisions that we cared about. It balanced the international reach for search warrants that the DOJ wanted with a recognition that tech companies could go to court to challenge warrants when there was a conflict of laws. This meant that if Ireland, Germany, or the entire European Union wanted to block unilateral foreign search warrants through their local laws and instead compel a more transparent or collaborative approach, they could do so and we could rely on this in a US courtroom. Even more important, the CLOUD Act created the new authority for modern international agreements that could replace these unilateral efforts. These agreements can enable law enforcement agencies to access data in another country with faster and more modern procedures, but with rules to protect privacy and other human rights.

Stelle: 1,475

We need to recall another consequence of technology that reinforces our tendency to splinter into cybertribes: being alone together. More and more, we find ourselves absorbed in electronic conversations with people we aren’t physically with. Sometimes we are a world apart. Digital technology has made the world smaller—and people more accessible—but it has also cast a deafening silence between people sitting next to each other.

Stelle: 1,641

There is likely to be room for complementary regulatory approaches, combining a more narrow focus on specific categories of objectionable content with a broader effort to provide users with more information on its sources. One important feature of the latter approach is its emphasis on addressing the spread of disinformation not by assessing whether content itself is true or false but instead by providing social media users with accurate information on people’s identity. It’s a commonsense approach that’s been adopted in modern-day political advertising. Leave it to the public to decide what is true. But let them make that decision based on an accurate understanding of who is speaking.

Stelle: 1,695

Until recently, communications technologies seemed to favor democracy and put authoritarianism on the defense. Now we must ask whether the internet has created an asymmetric technology risk for democracies that authoritarian governments can counteract

more readily than the republican form of government that Franklin's words urge us to protect.

Stelle: 1,711

While Denmark was the first country to name a formal ambassador to liaise with the tech sector, the country's decision followed a similar step by the British government. In 2014, Prime Minister David Cameron created a position in his office for a special diplomatic role, initially to address law enforcement technology issues and then to serve as "special envoy to US technology companies." The first to take the new post was Sir Nigel Sheinwald, a former British ambassador to the United States. Other governments from Australia to France have followed suit with similar moves. It's a shift that shows how the world has changed.

Stelle: 1,760

Just as governments had pledged in 1949 to protect civilians in times of war, perhaps a Digital Geneva Convention could capture people's imagination about the need for governments to protect civilians on the internet in times of peace. It was an idea that could build on work already in motion by governments, diplomats, and tech experts focused on establishing so-called cybersecurity norms between nations.

Stelle: 1,846

Governments will sometimes seek to evade international agreements if they can, so there need to be effective ways to monitor compliance and hold violators accountable. This speaks directly to one of the biggest challenges for controlling cyberweapons. Governments not only perceive them as useful but also uniquely easy to use in a manner that evades detection. As David Sanger of the New York Times has put it, this unfortunately makes them "the perfect weapon."

Stelle: 1,894

When we unveiled the Cybersecurity Tech Accord in April 2018, thirty-four companies signed on.²³ It was more than enough to generate momentum. By May 2019, the group had more than one hundred companies from more than twenty countries, and it was putting the accord into action by endorsing practical steps to strengthen cybersecurity protection.

Stelle: 1,973

Launched just a little more than six months apart, the Paris and Christchurch calls highlight the progress the world can make by advancing what Casper Klynge likes to call “techplomacy.” Instead of relying on governments alone, a new approach to multi-stakeholder diplomacy brings governments, civil society, and tech companies together.

Stelle: 2,154

The GDPR is akin to a Magna Carta for data. It represents a critical second wave of European privacy protection. The first wave came in 1995, with a privacy directive that required that websites notify consumers and get their consent before collecting and using their data. But as the internet exploded, people were inundated with privacy notices and had little time to read them. Recognizing this, Europe’s GDPR required that companies give consumers the practical ability to go online to view and control all the data that had been collected from them.

Stelle: 2,300

The integrated nature of the global economy and the long reach of Europe’s privacy rules will create pressure even on countries like China to adopt strong privacy measures. In other words, Europe is not just the birthplace of democracy and the cradle of privacy protection. It’s quite possibly the world’s best hope for privacy’s future.

Stelle: 2,562

Most people consider technology a product business. The industry’s products grab the public’s attention and shape the way we work and live. But in a world where today’s hits quickly become yesterday’s memories, a tech company is only as good as its next product. And its next product will only be as good as the people who make it. This means, in short, that technology is fundamentally a people business.

Stelle: 2,587

Private AI, a recent breakthrough that better protects people’s privacy by creating the technical capability to train AI algorithms on data sets that remained encrypted.

Stelle: 2,646

One tenet I always employed in negotiations was a simple one: Never let a negotiation narrow to a single issue that can produce only one winner, even if it means holding open some other topics on which agreement might seem in reach. Instead, broaden the discussion so that more issues are on the table. Create the opportunity for more give-and-take and a round of compromises that can produce a scenario that enables everyone to claim victory in the final stage. It was an approach that had proven indispensable to our ability to work through difficult antitrust and intellectual property disputes with governments and companies around the world.

Stelle: 3,010

At Microsoft in late 2015, Eric Horvitz raised these concerns within the computer science community. In a piece he coauthored in an academic journal, he acknowledged that most computer scientists regarded the apocalyptic risks of a singularity as remote at best, but he said that the time had come to look more seriously at a growing range of other issues.¹⁶ Satya picked up the mantle the next year, writing a piece in Slate suggesting that “the debate should be about the values instilled in the people and institutions creating this technology.”¹⁷ He offered some initial values that needed to be incorporated, including privacy, transparency, and accountability. By late 2017, we concluded that what we really were talking about was the need for a full-blown approach to ethics across AI. It was far from a simple proposition. As computers gained the ability to make decisions previously reserved for humans, virtually every ethical question for humanity was becoming an ethical question for computing. If millennia of debate among philosophers had not forged clear-cut and universal answers, then a consensus was not likely to emerge overnight simply because we needed to apply them to

computers.

Stelle: 3,021

At Microsoft, we ultimately settled on six ethical principles for the field. The first principle called on us to address the need for fairness, meaning the bias problem. We then moved to two other areas where there was already at least some public consensus—the importance of reliability and safety and the need for strong privacy and security.

Stelle: 3,028

Our fourth principle dealt with an issue that our employees had rallied behind since Satya became CEO in 2014. This was the importance of building inclusive technology that addressed the needs of people with disabilities.

Stelle: 3,033

While each of these four principles was important, we realized that they all rested on two other principles that were foundational to the others' success. The first was transparency. To us, this meant ensuring that information about how AI systems were making consequential decisions was public and understandable.

Stelle: 3,040

The final ethical principle of AI would be the bedrock for everything else: accountability. Will the world create a future in which computers remain accountable to people and in which the people who design these machines remain accountable to everyone else? This may be one of the defining questions of our generation.

Stelle: 3,044

AI-based decisions that impact people's rights in a significant way need to remain subject to meaningful human review and control. And this requires people who are trained to evaluate the decisions that AI is making.

Stelle: 3,136

The issues raised by AI touch on topics like the role of personal accountability, the importance of public transparency, concepts of individual privacy, and notions of fundamental fairness. How can the world converge on a singular approach to ethics for computers when it cannot agree on philosophical issues for people? This is a fundamental conundrum for the future. More than in the past this will require that those who create technology come not only from disciplines such as computer and data science but also from the social and natural sciences and humanities.

Stelle: 3,283

"I think leading a tech company is becoming more like leading a university," I said. "We have researchers with PhDs who are like the faculty. We have interns and young employees who sometimes have views similar to university students. Everyone wants to be heard, and some want us to boycott a government agency much like they want a university to boycott the purchase of stock in a company that's doing something objectionable." For me, there had been a couple of key takeaways from my trustee experience. Perhaps the most important was that well-intentioned students might not have all the right answers, but they might be asking the right questions. And these questions could lead to a better path that had eluded experts and senior leaders alike.

Stelle: 3,296

It became increasingly important to spell out when we thought it made sense for the company to take a position on a public issue and when we should not. We didn't view corporate leadership as a license to use the company's name to address any issue under the sun. There needed to be some vital connection to us. We felt our responsibility was fundamentally to address public issues that impacted our customers and their use of our technology, our employees both at work and in their community, and our business and the needs of our shareholders and partners. This didn't answer every question, but it provided a useful framework for discussions with our employees.

Stelle: 3,374

In July 2018, we published a list of questions that we thought needed to be considered 14 and asked people for advice about possible answers.

Stelle: 3,459

Facial recognition should be developed and used in a manner consistent with broadly held societal values. We published six principles corresponding to our legislative proposals, which we have gone on to apply to our facial-recognition technology, and we have created systems and tools to implement them.²⁷

Stelle: 3,609

Bill Gates has famously remarked, “We always overestimate the change that will occur in the next two years and underestimate the change that will occur in the next ten.”¹⁵

Stelle: 3,868

As University of Michigan professor Richard Nisbett noted in his 2003 book, *The Geography of Thought*,¹¹ these issues reflect differing and deep philosophical traditions that go back more than two thousand years. American thought is often based in part on philosophies developed in ancient Greece, while Chinese thinking is founded on the teachings of Confucius and his followers. Over two millennia these have emerged as two of the world’s most prevalent and influential—but also different—ways of thinking. While I’ve spent decades attending meetings around the world, Beijing stands alone as a capital where government discussions sometimes hark back explicitly to historical experiences that date back over two millennia. Or to be specific, to 221 BC, the year the Qin dynasty unified China. As Henry Kissinger has noted, China owes its millennial survival to “the community of values fostered among its population and its government of scholar-officials.”

Stelle: 3,877

The values that continue to guide official thinking in China today derive from teachings by Confucius, who died more than two centuries before the Qin dynasty was born. His teachings included a commitment to compassionate rule, devotion to learning, and a pursuit of harmony based on a hierarchical code of social conduct that included a fundamental duty to “Know thy place.”

Stelle: 3,884

As developed by Aristotle and Socrates, the very definition of happiness for the ancient Greeks “consisted of being able to exercise their powers in pursuit of excellence in a life free from constraints.”

Stelle: 4,096

People everywhere create data, and over the course of this century, it seems reasonable to expect nations everywhere to create data in some rough combination of their relative population size and economic activity. China and the United States may be early AI leaders. But China, as large as it is, accounts for only 18 percent of the world’s population.¹⁰ And the United States represents only 4.3 percent of the world’s people.¹¹ When it comes to the size of their economies, the United States and China have more of an advantage. The US represents 23 percent of the world’s GDP, while China represents 16 percent.

Stelle: 4,108

Given the nature and role of academic research, universities have begun to set up data depositories, where data can be shared for multiple uses. Microsoft Research is pursuing this data-sharing approach too, making available a collection of free data sets to advance research in areas such as natural language processing and computer vision, as well as in the physical and social sciences.

Stelle: 4,208

The Clinton campaign had relied upon its technical prowess and its early lead. The Trump campaign, by contrast and out of necessity, had relied on something closer to the shared-data approach

Stelle: 4,227

The open-data movement can take a page from open-source trends for software. At first that effort was hampered by questions about license rights. But over time standard open-source licenses emerged. We can expect similar efforts for data.

Stelle: 4,237

Current laws were mostly written before AI developments began to accelerate, and there are tensions between current laws and open data that deserve serious consideration. For example, European privacy laws focus on so-called purpose limitations that restrict the use of information only for the purpose specified when data was collected. But many times new opportunities emerge to share data in ways that will advance societal goals—like curing cancer. Fortunately, this law allows data to be repurposed when it is fair and compatible with the original purpose. Now there will be critical questions about how to interpret this provision.

Stelle: 4,271

Societally we should aim to make the effective use of data as accessible as electricity.

Stelle: 4,301

AI is unlike singular inventions from the past such as the automobile, the telephone, or even the personal computer. It behaves more like electricity in that it powers tools and devices that run almost every aspect of society and our lives. Like electricity, AI will run in the background and in many ways, we will forget it's even there, until the day the power goes out. Satya has dubbed this new reality "tech intensity," a term that describes the infusion of technology into the world around us.

Stelle: 4,397

Tech leaders may be chosen by boards of directors selected by shareholders, but they are not chosen by the public. Democratic countries should not cede the future to leaders the public did not elect.

Stelle: 4,398

All of this makes it important for governments to take a more active and assertive approach to regulating digital technology.

Stelle: 4,463

How can governments regulate a technology that is bigger than themselves? This is perhaps the single greatest conundrum confronting technology's regulatory future. But once you ask the question, one part of the answer becomes clear: Governments will need to work together.

Stelle: 4,479

Progress came not by dwelling on who was missing but on who could be persuaded to join. This in turn led to continued momentum and additional expansion later on.

Stelle: 4,484

Recently the world's democracies have become less healthy. Perhaps more than that of any group of societies, their long-term well-being requires new collaboration to manage technology and its impact.

Stelle: 4,504

The issues of the twenty-first century require initiatives that are both multilateral and multi-stakeholder in scope. One key to multi-stakeholder collaboration is the recognition of the respective roles that each group needs to play. Government officials play a unique leadership role, especially in democratic societies, as those elected by the people to make societal decisions. They alone have the authority and responsibility to chart the course of public education and make and apply the laws under which we all live. Companies and nonprofit groups can bring a civic spirit and complement and partner with governments, bringing additional resources, expertise, or data the public sector often needs. And companies and nongovernmental organizations can test new ideas by experimenting and moving faster, especially across borders. We all need to appreciate and respect each other's roles.

Stelle: 4,511

Many issues also will require compromise. For successful business leaders who have helped build some of the world's most valuable companies, this is not always easy to contemplate. They typically succeeded against long odds by doing things their own way; regulation will restrain their freedom in the future.